

RESEARCH EXPERIENCE

I build stable and trustworthy multimodal intelligence that unifies perception and generation across domains, modalities, and real-world settings. My research spans open-world and 3D perception, video/world generation, and AI for science, with the long-term goal of enabling general-purpose AI for digital life, autonomy, healthcare, and scientific discovery.

- AI for Digital Twins 2025 - Present
 - World Model; Digital Human
- AI for Autonomous Mobility 2020 - Present
 - 3D Occupancy Prediction; Open-vocabulary/Cross-domain/Open-set Object Detection
- AI for Scientific Innovation 2024 - 2025
 - Nanophotonics; Meta Optics; Graph-based Learning
- AI for Transforming Medicine 2020 - 2024
 - MRI Brain Analysis; Medical Report Generation; 4D Surgical Simulation

PROFILE HIGHLIGHTS

- Academic & Industrial Perspective
 - Founded a seed-funded Hong Kong startup while pursuing top-tier research (50+ papers; 15+ lead-author papers).
- Large-Scale GPU Experimentation
 - Extensive hands-on experience with large-scale GPU clusters (10,000+ GH200).
- Leadership & Collaboration
 - Led a Swiss national AI initiative with 15 leading PIs; NeurIPS workshop organizer.
- Rapid Citation Impact
 - U-KAN (co-first author) received 600+ citations within 1.5 years.
- Impactful First-Author Project
 - One first-author project attracted 2.4k+ GitHub stars and was deployed on the Poe platform.

RESEARCH HIGHLIGHTS

- Selected as Oral/Spotlight/Highlight Presentation (1st Author)
 - AAAI 2022; CVPR 2022; ICCV 2023; ECCV 2024; ICCV 2025; NeurIPS 2025; ICLR 2026
- Selected as Best Paper Finalist (1st Author)
 - CVPR 2022 (33/8161)
- Full-Score Submission (1st Author)
 - ICCV 2025, final scores: 6, 6, 6
- Selected as Cover Paper (2nd Author)
 - ACS Photonics, 2024
- AAAI 2025 Top-1 Influential Paper (Co-1st Author)
 - U-KAN: medical image segmentation and generation backbone

WORK EXPERIENCE

- Ant Group Jun. 16, 2026 - Present
 - Staff Research Engineer
 - Research areas: multimodal intelligence, world models, and AI for digital twins
- École Polytechnique Fédérale de Lausanne (EPFL) Feb. 2025 - Jun. 2026
 - Research Scientist, Robotics and Computer Vision
 - Supervisor: Prof. Alexandre Alahi (Advised by Prof. Feifei Li@Stanford)
- The Chinese University of Hong Kong (CUHK) Mar. 2024 - Feb. 2025
 - Postdoc, Electronic Engineering
 - Supervisor: Prof. Yixuan Yuan
- ScholaGO Education Technology Company Limited, Hong Kong 2024 - Present
 - Founder & CEO
 - Developed multimodal educational products; secured seed funding from HK Tech 300 and HKSTP.

EDUCATION

- City University of Hong Kong (CityU) Sep. 2020 - Feb. 2024
 - Ph.D., Electrical Engineering
 - 3.5-Year; Early Graduation
 - Supervisor: Prof. Yixuan Yuan
- Tianjin University (TJU) Sep. 2016 - Jun. 2020
 - B.S., Electrical Engineering
 - GPA: 3.83/4.00 (Top 2%)

1. **[Digital Human] EverAnimate: Minute-Scale Human Animation via Latent Flow Restoration**
Wuyang Li, Yang Gao, Mariam Hassan, Lan Feng, Wentao Pan, Po-Chien Luan, Alexandre Alahi
 Neural Information Processing Systems (NeurIPS), 2026
2. **[World Model] SurgVista: Long-Horizon Surgical World Modeling with Plausible Instrument-Tissue Dynamics**
 Wentao Pan*, **Wuyang Li**^{†,*}, Shengyuan Liu, Xinyu Liu, Hengyu Liu, Yixuan Yuan
 Neural Information Processing Systems (NeurIPS), 2026
3. **[World Model] Drift-Resistant Navigation World Model with Anchored Epipolar Guidance**
 Po-Chien Luan, Zimin Xia, **Wuyang Li**, Yang Gao, Alexandre Alahi
 Neural Information Processing Systems (NeurIPS), 2026
4. **[Video Generation] Proprio: Latent Self-Scoring and Inference-Time Refinement for Physically Plausible Video Generation**
 Mariam Hassan, Kaouther Messaoud, **Wuyang Li**, Alexandre Alahi
 Neural Information Processing Systems (NeurIPS), 2026
5. **[Video Generation] Anchored Video Generation: Decoupling Scene Construction and Temporal Synthesis in Text-to-Video Diffusion Models**
 Mariam Hassan, Bastien Van Delft, **Wuyang Li**[†], Alexandre Alahi
 Under Review, 2026, High-score ECCV submission
6. **[Trajectory Forecasting] Social-Mamba: Socially-Aware Trajectory Forecasting with State-Space Models**
 Po-Chien Luan, **Wuyang Li**[†], Yang Gao, Alexandre Alahi
 European Conference on Computer Vision (ECCV), 2026
7. **[Video Generation] Stable Video Infinity: Infinite-Length Video Generation with Error Recycling**
Wuyang Li, Wentao Pan, Po-Chien Luan, Yang Gao, Alexandre Alahi
 International Conference on Learning Representations (ICLR), 2026, **Oral (1.3%)**
8. **[3D Perception] Deformable Gaussian Occupancy: Decoupling Rigid and Nonrigid Motion with Factorized Distillation**
 Yang Gao, **Wuyang Li**[†], Po-Chien Luan, Alexandre Alahi
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026
9. **[3D Perception] VoxDet: Rethinking 3D Semantic Occupancy Prediction as Dense Object Detection**
Wuyang Li, Zhu Yu, Alexandre Alahi
 Neural Information Processing Systems (NeurIPS), 2025, **Spotlight (3%)**
10. **[AI for Science] MetaScope: Optics-Driven Neural Network for Ultra-Micro Metalens Endoscopy**
Wuyang Li^{*}, Wentao Pan*, Xiaoyuan Liu*, Zhendong Luo, Chenxin Li, Hengyu Liu, Din Ping Tsai, Mu Ku Chen, Yixuan Yuan
 IEEE International Conference on Computer Vision (ICCV), 2025, **Full final scores (6, 6, 6)**
11. **[Backbone] U-KAN Makes a Strong Backbone for Medical Image Segmentation and Generation**
 Chenxin Li*, Xinyu Liu*, **Wuyang Li**^{*}, Cheng Wang*, Hengyu Liu, Yifan Liu, Zhen Chen, Yixuan Yuan
 Association for the Advancement of Artificial Intelligence (AAAI), 2025, **AAAI 2025 Top-1 Influential Paper**
12. **[2D Perception] CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection**
Wuyang Li, Xinyu Liu, Jiayi Ma, Yixuan Yuan
 European Conference on Computer Vision (ECCV), 2024, **Oral (1.8%)**
13. **[Video Generation] Boosting Medical Image Analysis via Motion-Informed Generative Videos**
Wuyang Li, Xinyu Liu, Qiushi Yang, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
14. **[Open-World Learning] Novel Scenes & Classes: Towards Adaptive Open-set Object Detection**
Wuyang Li, Xiaoqing Guo, Yixuan Yuan
 IEEE International Conference on Computer Vision (ICCV), 2023, **Oral (1.8%)**
15. **[Open-World Learning] Adjustment and Alignment for Unbiased Open Set Domain Adaptation**
Wuyang Li, Jie Liu, Bo Han, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023
16. **[Domain Shift] SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2023

17. **[Domain Shift] SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022, **Best Paper Finalist (0.4%)**
18. **[Domain Shift] SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation**
Wuyang Li, Xinyu Liu, Xiwen Yao, Yixuan Yuan
Association for the Advancement of Artificial Intelligence (AAAI), 2022, **Oral (4.5%)**
19. **[Brain Analysis] Synthesizing Realistic fMRI: A Physiological Dynamics-Driven Hierarchical Diffusion Model for Efficient fMRI Acquisition**
Yufan Hu, Yu Jiang, Wuyang Li, Yixuan Yuan
International Conference on Learning Representations (ICLR), 2025
20. **[Brain Analysis] InstantSplamp: Fast and Generalizable Steganography Framework for Generative Gaussian Splatting**
Chenxin Li*, Hengyu Liu*, Zhiwen Fan, Wuyang Li, Yifan Liu, Panwang Pan, Yixuan Yuan
International Conference on Learning Representations (ICLR), 2025
21. **[Domain Shift] Universal Domain Adaptive Object Detection via Dual Probabilistic Alignment**
Yuanfan Zheng*, Jinlin Wu*, Wuyang Li, Zhen Chen
Association for the Advancement of Artificial Intelligence (AAAI), 2025
22. **[AI for Science] Stereo Vision Meta-Lens-Assisted Driving Vision**
Xiaoyuan Liu, Wuyang Li, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen
ACS Photonics, 2024, **Selected as Cover Paper**
23. **[SAM] Flaws can be Applause: Unleashing Potential of Segmenting Ambiguous Objects in SAM**
Chenxin Li*, Yuzhi Huang*, Wuyang Li, Hengyu Liu, Xinyu Liu, Qing Xu, Zhen Chen, Yue Huang, Yixuan Yuan
Conference on Neural Information Processing Systems (NeurIPS), 2024
24. **[SAM] When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels**
Yifan Liu, Wuyang Li, Cheng Wang, Hui Chen, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024, **Early Accept (14.0%)**
25. **[Diffusion] Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation**
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
26. **[Brain] fMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction**
Zhibin He, Wuyang Li, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
27. **[3D] LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction**
Hengyu Liu*, Yifan Liu*, Chenxin Li*, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
28. **[Video Generation] Endora: Video Generation Models as Endoscopy Simulators**
Chenxin Li, Hengyu Liu, Yifan Liu, Brandon Feng, Wuyang Li, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
29. **[Brain] Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer**
Zhibin He, Wuyang Li, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan
IEEE Transactions on Medical Imaging (TMI), 2024
30. **[Multi-Modality] MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics**
Qiushi Yang, Wuyang Li, Baopu Li, Yixuan Yuan
IEEE International Conference on Computer Vision (ICCV), 2023
31. **[Noisy Labels] Intervention & Interaction Federated Abnormality Detection with Noisy Clients**
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022, **Oral, Early Accept**
32. **[Noisy Labels] Towards Robust Adaptive Object Detection under Noisy Annotations**
Xinyu Liu, Wuyang Li, Qiushi Yang, Baopu Li, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022
33. **[Object Detection] HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection**
Wuyang Li, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan
IEEE Transactions on Image Processing (TIP), 2021

AWARDS

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|---|-------------|
| • Outstanding Academic Performance Award, CityU | 2022 - 2023 |
| • Outstanding research achievements | |
| • Research Tuition Scholarship, CityU | 2022 - 2023 |
| • Outstanding research achievements | |
| • Ranked First in Polyp Detection Challenge | 2021 |
| • 3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021) | |
| • Undergraduate National Scholarship | 2017 - 2018 |
| • The highest undergraduate honor for a student in China | |
| • Silver Award in Tianjin Advanced Mathematical Competition | 2017 |
| • As a selected participant representing Tianjin University | |

SERVICE

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| • Leader of Large Swiss AI Initiative Project [Link] | 2025 |
| • Led a national project with 15 leading Swiss PIs; gained large-scale GPU cluster experience (10,000+ GH200). | |
| • NeurIPS 2024 Workshop Organizer [Link] | 2024 |
| • Organized the Advancements in Multimodal Medical Foundation Models workshop | |
| • Research Funding Proposals | 2022 - 2024 |
| • Assisted my advisor in preparing several competitive research funding proposals | |
| • Startup Funding [Link] | 2023 |
| • Secured seed funding from HK Tech 300 and HKSTP | |
| • Founder, ScholaGO Education Technology Company Limited, Hong Kong [Link] | 2024 |
| • Founded a company developing multimodal educational products | |
| • Conference and Journal Reviewer | |
| • ICLR/ICML/NeurIPS; CVPR/ICCV; TPAMI; IJCV; TIP; NeurIPS Top Reviewer | |